

ActInf Textbook Group ~ Cohort 7 ~ Session 4 (Chapter 2, part 2) ~ 8/12/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 4 (Chapter 2, part 2) ~ 8/12/2024 | 1:01:20

all right welcome back cohort 7 we're in
our second discussion on chapter two so
let's head over to
two we can begin with
anyone's thoughts or Reflections or
any question or idea or section of the
text they want to begin with OR
curiosity from to or just some quote
they want to go to
anyone can just raise their hand or type
it in the
chat okay actual go for it sure so one
thing that I found a bit interesting
it's on page 26 of the book um it says
the person can resolve this discrepancy
in two ways it's talking about this
thing of looking at um like expecting to
see an apple when you look and then
actually seeing a frog and it says first
she could change her mind about what
she's seeing so actually that it's a
frog instead of an apple um or she can
fiat the nearest apple tree and see
something that looks very much like an
apple this resolve the initial
discrepancy but in a different way um so
she's kind of doing an action to make
the world fit more her model and they
they do another example afterwards but I
I'm kind of curious about exactly how
this tradeoff is done when would you
choose to do one thing over the other
when would you choose to do an action
versus when would you choose to do a um
like an update of your internal model uh

especially in a situation where you don't know you know perhaps if you've had this situation happen many times you kind of learned that it's important to know that it's a frog uh so you don't want to look away you want to up your model but what if you're meeting it the first time I was a little bit confused about

that great question anyone have a thought on

this you're likely doing both in parallel much of the time somebody who's too fixated on one or the other that's tends to be a neurotic problem or Worse what that's greater um but there a few few ways to take

this first there's the secret third option which is neither discrepancy doesn't have to be resolved but it is kind of like the equivalent of a ball on a shelf it's like an unresolved potential energy or like an unburnt candle it's like an unresolved Gibbs free energy differential however it isn't the case that a discrepancy is resolved one of these two ways however these are the two ways that there can be the reduction or of discrepancy or or discrepancy continue to increase so then the question is like okay in in what ways uh are or could be discrepancy resolved through inference and through through action

um there's defin ways to take this um in the in the particular partition so that's splitting up into these four nodes these two nodes that constitute

the the two directions of the blanket
and then the internal and the external
states that are partitioned off from
each other from that blanket those are
all happening
simultaneously so one answer is they're
both separately in play so you could
have a situation where where neither uh
belief updating nor action updating were
happening like a rock you could have a
situation where both were
concurrently um
updating sometimes it can be helpful to
think about these kind of multiple time
scales of what changing the mind can
look like the fastest time scale which
is kind of equivalent to the forward
pass on a neural network is just the the
forward inference that's just all the
kind of parameters are staying the same
and then new observation comes in basy
and update cranks then at a slightly
slower time scales um corresponding to
attention and learning mechanisms
there's kind of like those are
corresponding to like weight updating in
a neural network and so those those are
like slower kinds of changing the
mind um changing the world through
action also like that's very dependent
on what the kind of action is for
example
here the um eyes were moved so it's a
muscular action to bring the
observations into alignment with the
expectations however there wasn't a
change in the world so visual activity
is kind of a special kind of action

because it doesn't modify the world but it is still a way to reduce discrepancy through action so it's kind of a kind of on the borderline like is that changing the world me your eye is part of the world so that is but it it doesn't modify like any kind of latent State then there are the actions that have like a more you know turning on the air conditioner or the heater that actually modify the world so short answer they're both happening uncoupled and then when you look at the specifics of the generative model then you can you could separately talk about like the probabilities of a model to update its beliefs if it has a fixed belief distribution then it's not going to update its beliefs if it has a very broad belief distribution prior then observations may update its beliefs a lot and then similarly depending on like the efficacy and the agency of the actions that are being discussed then maybe taking actions to modify you know changing the temperature in the room if you have a really fixed physiological prior on body temperature and the air conditioner is right there then that's going to be reflected in the model and you'll see changing the world if there's no air conditioner or the agent doesn't know about it and it's able to like learn different physiological um temperatures or has a very broad temperature range that it could be okay

in maybe that would be the behavior that you'd see here's like the temperature simply changing one's mind does not seem very adaptive but acting to lower the body temperature is right thanks a lot and I think I was just interested in about um so that makes sense to us that it uh that we get more out of CH trying to actively change our body temperature than just believing oh um I actually don't have this body temperature anymore but where are we learning that is that through some kind of trial and error that we're learning this is this like an evolutionary prior or where did we get this from yeah it's a it's a good question so um when you construct the model then you get to create whatever the distribution is then ask you well but how how did that prior on body temperature come to be and that the evolutionary account is um you know in the 98° thermal bacterial niche the bacteria that didn't have optimal growth that didn't expect themselves to be at that temperature those have been swept off the table so we end up seeing a phenotypic distribution that's the the fitness distribution of the bacteria in the thermal vent that corresponds like to a thermal gradient um another another kind of consideration is actions are being selected also from a probability distribution like there's

no proposal of some auxiliary reward
distribution we're updating our policy
prior habit into the policy posterior
we'll look at that like later in the
chapter so we're doing inference on
action we have inference on perception
those the that's active inference um so
in terms of the belief Distribution on
perceptions that
you can think okay like the beliefs
about the temperature in the room but
also there's the beliefs about turning
on the air conditioner so now which
which um then the the um sort of
liability of both is defined so if the
distribution that one of these if the
Habit distribution is like um 100% I
turn it on 0% heater you know 100% air
conditioner 0% heater then that is like
it's like terministic action selection
versus one that's more spread out across
the action possibilities so you can kind
of get that interpretability when you
see the model and then that's why
there's like when you do specify model
it's like and this comes up in chapter
six you kind of think more structurally
about the
model and then there tends to be a lot
of parameter sweeping and like
investigation across different
combinations like you could try
10 different um kind of widths of Prior
on the perception side from very tight
to very loose and then similarly on the
action side sweep across 10
by1 and then look at um you know color
in each cell by the probability of times

that they turned on the air conditioner so it's it's not always just about finding like one kind of specific parameterization especially with these more intuition pump models but often it's more about sweeping and then looking at like oh um you know there's kind of a there's a manifold where on this side this is the case and on that side this is the case thanks a lot yeah thank you um any other question or chat oh also just to sort of yeah yeah go ahead

Andrew are you saying this but the space could potentially be not smooth right if there are if there are Jagged jumps in terms of response to different probabilities does

does I'm not trying to be too technical but does that make sense it it could be a a jagged space that has like piecewise squares or something yeah it could be a fully discrete space I mean there there could be a continuous space Also with with nonlinearities but also it could just be a fully nonlinear or I mean a fully discrete space like it could be a checker game or a chess game so then that then that um then those sweeps would have sharp boundaries okay

look's yeah go ahead oh that just matches

that it doesn't have to be nice cuz I was thinking why would you do parameter sweeping instead of some type of optimization like but there may not

be a nice you know smooth space to do optimization

on

so yeah yeah I mean you you've explored

that and know a lot of areas sometimes

when you fully like the fully synthetic

when you fully get to specify it like

we're playing Connect Four so it's a

discrete space there's a discrete action

space there's a discrete game playay

space or

tic-tac-toe then it's like well behaved

and discret but then when you get into

like

discretized forms of systems that have

continuous character

then like then there's all these other

issues that could come

up also just want to make one kind of

General comment just chapters one and 10

are are like book ends it's like

introduction and kind of conclusion and

they're they're similar chapters they

contextualize active inference chapter

four is where we're going to meet the

actual generative model itself the

generative model is kind of like the

transfer station or the midpoint on the

high road and the low road we're in

chapter two so we're talking about model

architecture so these are things that

you can build these are ba go starting

from Bas

equation and using basian statistics to

build up architectures that have a

perception inbound sense making element

and an act

outgoing control theoretic

element the high road is where we'll get to the free energy principle it's kind of like imperative for survival and some of those topics but in the low road it's kind of like tinkering and cobbling with different model architectures and motifs I have a question uhhuh so I'm from the transitioning from uh free energy to expected free energy um I mean I I guess it makes a lot of sense to to actually take the expected value of the the free energy but is there like that derivation wasn't really that clear to me it was just like here's the expected free energy do you know have any good resources or where you can actually see that derivation great question well one is in octopus's math

work um

I I think tomorrow and and continuing it'll be drawn out um a little bit also over it you can and you can follow like click on the little icon for the a here's equation 2.5 so few few things about the equation um we have the screenshot also there's there's the uh plane

latac this can help if you are putting it into like a language model and we have some of these experiments with interpreting and asking questions about it not not saying they're all 100% like complete or anything but in the canvas for two point and then also for for many but not all the equations there's the um natural

language pronunciation basically which

can be helpful and so if it's missing
people people can add it so that that
can
um we we'll have to fix this because it
looks like a lot oh I I kind I know what
happened here
okay there we go so these are the lines
all right in 2.5 there's a few resources
so
um let's
just where do how do we get to 2.5 and
2.6 these are some of the key equations
okay 2.5 equation 2.5 is variational
free energy it these are both v_f and E_F
are functionals so there are functions
that have the arguments that are other
functions that the arguments can include
other functions so f is the functional Q
is the variational belief distribution Q
is the distribution that we get to
control the form of so it is set by the
model to have like an analytically
tractable form like a gaussian
approximation so even if the world is
non you know the world's biodal and then
the model saying but but I'm using a
gaussian so that gives me all these like
nice methods then also all the
pathologies like if the world's biotal
and you're fitting a gaussian you could
you know potentially you fit to only one
hump but not the other or you end up
blurring over both the middle so it's
not not not that if you know if there's
a mismatch on form then this will lead
you downhill to not necessarily like the
best of all possible outcomes but given
the selection of the form of Q it will

be the best possible outcome it's a functional of the belief distribution and one incoming data point why so variational free energy is kind of like this real time check on like how surprised am I with this data point variational free energy bounds surprise minimizing surprise is equivalent to maximizing model evidence if you were minimally surprised with the new temperature readings coming in that would be equivalent to saying you had like the maximum likelihood model of the temperatures coming in you're making the best possible predictions is equivalent to saying that you were minimally surprised so this is bounding minimal surprise then um it is only dealing with belief in a moment and one data point coming in so that's why it's this kind of like real time sense

making

indicator

2.6 is when we start looking into talking about observations that haven't happened yet at the future and action planning G expected free energy it's taking in π which is this policy vector it sums to one it's a probability distribution over actions the policy prior is the Habit distribution G is going to sharpen that policy prior into a policy posterior which then you could sample from like according to the probabilities or you could just select the likeliest action okay how do we get between the different lines again in this canvas here for 2.5

and 2.6 there's some useful resources um
um here's equation 2.5 walk
through um Ali
ramju Yak smal Jonathan shock have all
done like have have contributed
variously to these sections so that's
one good

PDF

um here I

think this was

yakups the interesting thing for 2.5 and
and this is like again why there's so
many dots to connect with the math is
2.5 actually um it's not
that one you go to two and two you go to
three maybe there's a maybe there's a
way to do that as well but here's the
derivation going from one to two and
then one to
three so these may be some useful
resources

also

here let's see if this resource is still
there Professor Jonathan shock has
written up a sort of math
overview August 12 I don't know if that
renders to the current date just or if
it was like actually edited at this
current date that also has a lot of
information so that's something that
that takes a lot of uh coats of paint I
think but it it really is there and
grounded and if we can pull out and help
highlight

like all those threads and show okay
here's where this is grounded in the
definition of Shame and entropy or
definition of surprise and likelihood

that'll be

cool one other thing I'll say is um

variational free energy is kind of more

of a classic quantity it's the quantity

that's used in variational inference um

like variational physics variational

Auto encoders and so on whereas expected

free energy is just one choice of a way

to model

futures for example there's also this is

EF expected free energy also there's

free energy of the expected future so

there's several other future oriented

approximators that have been proposed

but variational free energy is kind of

the

classic

Adam yeah like something that I'm

wondering about and I haven't really

quite like understood yet um is that so

from like my my own personal past and

like where I'm coming from up until

active inference I had been reading

quite a bit about like maximum entropy

inference in like John Hart from

Berkeley and Edward Thomas James and uh

even like pier

llas and um trying to think so anyways

definitely like ET James he was really

like a proponent of Max ant inference

and um

prior to active inference I had I had

been thinking about like the stability

of syst systems in the maximum entropy

State and the way

that if if the energy of the of the free

energy equation is is constant then by

the increase of entropy it automatically

minimizes the free the free energy right
so just that the Min the men free energy
is the equivalent
or not necessarily but can be equivalent
to the maximum entropy and
so I don't know if anyone else has been
thinking about this or maybe if if you
like also I don't know any thoughts on
that yeah that's a great question it it
H has not been um fully fully fleshed
out but here's one kind of
recent discussion where it came up and
where they explicit ly highlighted
this okay so here was it it
um okay this is the octave data sampling
paper so
observations um it's kind of this is
kind of like like building on that
discussion of the equation 2.5 VF so
that's where the variational free energy
for data point and a prior are going to
be
described you could also look at a data
point and and your prior and disc like
what was the epistemic value of that
data point coming in how much did we
learn from it um here they're in a
clinical study setting so like the real
Target uncertainty to resolve is about
the efficacy of these different
treatments
um they use max ENT sampling
and then they also compare that with the
expected Information Gain driven
sampling
so there's a very
um okay here we go several examples
would yield identical results had we

adopted data sampling based upon Max entropy sampling the key differences emerge when variance around predicted outcomes is inhomogeneous so when there's equal variability profiles like let's just say that we're we're interested in the temperature distribution in this room and all of our sensors have equal VAR sensor variability then we can sample from the most the sections that we're least certain about the temperature and that's like a Max ENT strategy however let's just say that there was regions in the room that had different variability and there were sensors with different variability so just to kind of simplify let's just say we had high and low variability sensors and there's high and low variability areas in the room so if we were just going to sample based upon variability period measurement variability like total variability then maybe this we'd be ending up sampling from sensors that were highly variable but regions where we actually were not not that uncertain about the temperature at that location so in the case where the sensors where the sensor variability we don't have to worry about then these approaches are equivalence but there's actually an additional level of nuance possible so it's it's closely aligned with Max and I

think again it would be it would be
great to to draw this out better
but
um it's sampling from a total epistemic
value which is able to kind of
like accommodate that case where there's
variability sensors and the variability
of the temperature parts of the
rim yeah I see that I'll have to look
into that that sounds interesting cool
yeah the the slides are there the and
the paper um and and we we that's an
interesting thing to keep on
exploring what exactly what exactly um
this is just on the website I can get to
this thep yeah um just here's the just
like 57.0 that was the the background
and then 57.1 and point2 um Carl friston
and Thomas Parr
joined I forget which one of the weather
was d one or. two but in one of them
Carl gave a a very vifer defense of
epistemic value because often times
people say Okay epistemic value learning
in service of pragmatic value right like
learning is good for something when it's
pragmatic and he gives a extended
thought on like how they're kind of
separate but they're unified under the
expected free energy but um like
epistemic value is its own distinct kind
of value from pragmatic
value whether or not those learnings
contribute to epistemic or I mean
contribute to pragmatic
outcomes I
don't no sorry all over my hand okay
yeah

th this the epistemic um like active
data sampling like on the biological
side it's really interesting because it
describes for example is
secades epistemic driven action
selection but then also they explore
having a cost function so that could be
like how much how much cents does it
cost to make an API
call and then that is the
parameterization question is like how
much epistemic value how many bits of
information are worth one cent of an API
call um or in the case of the clinical
trial there's like you could have a cost
associated with the number of times that
sampling occurs or um cost reflected by
a preference for survival but if the
preference is purely for survival and it
was like some kind of extreme utility
driven clinical trial then a sampling
algorithm would choose to sample nothing
so it' be uninformative with zero
observations however it would have
achieved maximum preference satisfaction
because it wouldn't have observed any
sickness or
death so they talk about a few of these
kind of like interesting
cases and they they build it up very
clearly like how do you add in the cost
and all these different
things and this is like a different
graphical model than comes up in the
textbook again it's the clinical trials
setting but but to connect it with the
textbook
these are basing graphical

models and so a lot of the the kind of structural analysis of a situation comes down to what are the variables what are like the observable variables what are the environmental variables what are the and there's there's not really like an agent like kind of sense an action agent unless you chose to see like the clinical design team's observations and decisions in in that light um what are the observations what are the environmental variables what are their influences upon each other that lays out like the skeleton and then there's the question of either generating synthetic data from that skeleton in that in the forward direction or explore in chapter n of the textbook is like how do you go from observations to inference on those latent factors like um what is the efficacy of the treatment anyone else give like a thought on chapter 2 or question or a page or we can look at some of the questions come on go for it yeah thanks um yeah I've been thinking a bit about like the distinction between pre energy and expected pre energy and sort of thinking about like so in like in a way aren't all actions requires planning or like is there actually a clean distinction in practice when we when we minimize F and when we minimize G or is it like a bit of a fussy line in between and in

practice we're only doing the one or if
you know what I'm getting at
yeah great great question so let let's
go to pmdp documentation that this can
this can um clarify a few of those
things

one uh heris stic is if you're
considering explicit planning so more
than one time Step Ahead like policy
sequences that have a Time Horizon
that's not

one if you want to do explicit planning
Tre search policy roll out um deep
active

inference then you do need to use
expected free energy so if explicit
planning is going to come into play then
you go with expected free energy if it's
just

single time step then it can you you can
just sort of have that reactive
updating um yeah hon let me um that's in
in the

uh
equations equation 2.5

put in the
chat yeah

so if there's explicit planning yes EF
has to come into
play if you have an agent that is not
doing explicit

planning it could do action selection
using vfe so what that would look like
and and is like you have um a belief
Distribution on the temperature in the
room and a belief Distribution on your
habits related to the air condition
conditioner like I'm 50% likely to turn

on the air conditioner or not new
observation comes in it instantaneously
updates the belief distribution that's
the sensemaking element and it
instantaneously updates your um your
policy distribution which is then
sampled from so there's not roll out or
anticipation explicitly of future
unobserved outcomes which is what you
would get with
ef but you only get one time step there
um here in in
pmdp it's
um they really especially in the um
documentation make it pretty clear and
declarative like there's only two
actions here but here even for a single
time
step moving left and moving right
they're they are considered with
expected free energies because there's
actually like a question of what what
would I expect to see if I went left and
what would I expect to see if I went
right so that's kind of like you know
there's a lot of way ways to shade that
but that's kind of like the theological
aspect of action is when expected free
energy comes into play which is like
what what would I see if I went left
versus what would I see if I went right
right versus the sort of past pushing
forward without an explicit teleology
would be like I have a 50% chance of
left and right prior on policy I just
saw um a light or a glucose gradient
from the left that just bumped me to
6040 I sample from the 6040 and I go

left but it wasn't based around what
would I see so it wasn't like based
around that what what is the epistemic
value and the pragmatic value
of these two

Alternatives that's the
real policy like planning as
inference versus more of like a
reactive policy just habit unupdated or
updated habit distribution and then just
drawing from that without explicit
consideration of the
consequences and maybe just a short
followup question

um so yeah I guess you can make it like
really clear when when writing a
program about it but in the context of
of living organisms when you yeah so
like what is what is one time step then
um could maybe

like develop that discussion a bit on
like specific for living organisms

[Music]

um yeah well math territory H how do we
develop continuous Andor discrete time
models maps of complex multiscale
systems like there's not really a
natural time click for a body per se
it's not simply like the heartbeat it's
not the seconds it's not the breath so
then there's a lot of Moder degrees of
freedom with how they
model time

um and then to what extent the modeler
chooses to test here's some models I
made where there's planning happening
and here's some models where planning
isn't

happening that that that's one thought
but but yeah
like does anything plan or when does
planning happen or like one old active
joke was like active inference only
works for one time step
but then you can compress and make that
one time step very
complex so that you like it could be one
mult like one nested time
step that that would subsume what looks
planning like if if um let's just say
that we have we were um talking about
like financial decision making or
something like that and we had a a a
policy decision that was being made at
the yearly time
scale if someone only knew about
day-to-day decision making that one time
scale of the yearly model might look
like it had been planned over many
days or there could be a map that just
said we have a single policy we make one
decision per
year but then that model wouldn't be
able to make decisions day by day but it
might look you know it this just gets
into like how I mean how and we can make
many models from the biological
phenomena and then that that's kind of
the interesting question like is there
evidence for planning in is
secades or is it simply
marcovia and we can totally account for
is secades based upon location of the
Gaye at a moment the incoming
observation and the belief distribution
about what would be epistemically

valuable or like is there a
residual that that model can is not
accounting for that is suggestive of
like twostep Cade Planet I don't know
but those are like the kinds of of
research Pathways where where if you are
looking for where is their planning
happening thank you

um

actually sure so uh I'm pretty new to
active inference and I'm kind of trying
to right now get a intuitive feeling for
some of the um equations especially the
one we were looking at before I'm
wondering if we could just talk a little
bit about how the terms kind of relate
to each other and how actually I might
be able to send you uh I just kind of
cooked up something quickly before here
in the chat I kind of tried to create a
visualization of the um I don't know if
you have access to it uh I sent the link
Yeah so basically let's see here yeah so
the idea is that you have like the Q of
 X and the P of X and Y and as far as I
understand I might have made some
mistakes but as far as I understand you
minimize the free energy we're looking
at like the top uh version of the
equation right the first line um so we
have the Q of x which is our belief
distribution is that correct like our
belief of the Hidden States yep then we
have the P of Y and X which is just the
probability of Y and X happening as far
as I understand when P of y and x and q
of X are like exactly the same that
minimizes free energy is that

correct yeah well great work with this
this is this is super cool and it'd be
awesome to to continue to develop these
like interactive slider based things
which which many people have kind of
like brought up but um yeah like what
variational optimization does is the
modeler has chosen their family of
variational distributions that they're
going to be sliding the well- behaved
knobs on so like we're going to be using
a gaussian again that's not saying the
world is gaussian it's saying the the
modeler is choosing to to do
optimization you know descent with this
G um and then the ball rolling downhill
on that variational free energy
landscape until it rests at the bottom
what is the bottom of that energy well
it's where that observation coming in
given the belief distribution is is the
least surprising possible or it it
bounds surprise now that bound if the
generative process were gaussian and the
generative models gaussian
then you'll get like then it to it bound
surprise and and kisses it if the
generative process were something
different and you were doing a gine
approximation you still would get the
best possible gussian
approximator but then there would be
still like a
constant that that was like surprised
that couldn't be reduced because you
were fitting the best possible Galaxy
but it still was being
um because it was like not the same

family as a generative
process couldn't reduce all the
uncertainty but yeah this is really cool
right like this is just a very quick um
but but I do think it helps to kind of
play around with it and get a feeling
for how things change because one thing
I read in the book was that if you set
the mean of both to zero um you can see
that uh have to add some kind of quick
reset button as well but you can see
that uh as the they say that if you have
little evidence which would mean that
your P of X and Y would be really really
um kind of broad right so if you turn
the the bottom uh all the way up like
the yeah you you turn it up up up up up
then you start getting a free energy
that's higher but then actually as you
they say that you should as far as I
understand it that when you don't have a
lot of evidence you should also be very
uncertain certain so if you I'm seeing
the P of uh X and Y distribution as
being broad as not having a lot of
evidence and then if you take the Q of X
and take the standard deviation up on
that while the uh or sorry the standard
deviation you'll see that if the
standard deviation of the P_X and Y is
also very very broad it does
minimize uh variational free energy as
well does that uh make sense with what
what the truth is
or yeah like I'm I'm really interested
in this um change from a
uh single
well to potentially I mean potentially

some

plateau and then here potentially like a symmetry break where if the first observation coming in were here maybe it breaks to the right and then continues to follow the gradient and down to here and then a future observation couldn't even if it came from here it'd be like okay now back to here and back so then it's like trapped in an energy well so it's like what are these situations where even when we have a gaussian prior and a gaussian generative process like what is happening where we're we're getting a more like a um quartic like kind of two energy wells free energy distribution so I'm not sure if I'm illustrating it in the wrong way and maybe uh like the blue graph is basically Q of x times the natural logarithm of P of Y and X before it gets kind of summed up into the so that's kind of what that's showing I don't know if that was the same thing you were referring to but just so you know yeah cool maybe octopus and others can can play with it and keep keep on developing it or um or post the the repo that it comes from but this looks cool um so just one quick followup so earlier in the beginning you were talking about I mean can I understand it in that way that when you're very certain because they say in the book that if you are if you have little evidence then you should also not be so certain in your model um and I'm wondering whether that uncertainty is characterized by having a

very broad Goa for example and whether the lack of evidence is also uh kind of how do you say represented by having a very broad gion if you're using a go

yes this kind of relates like you have to the standard deviation um what comes up later is this like learning by counting so let's just say we've observed one heads and one Tails we might have um a distribution belief what is the probability of this coin coming up heads it's centered at 0.5 but it might be very broad whereas if we had one million heads and one million Tails observed then we would have very sharp Gan around 0.5 so there's different ways to kind of implement that but but broadly when you're less confident when there's more epistemic value on the table then belief distributions come to look like

uniformity and then as you gain Precision it gets more of a central tendency and then the extreme case is basically a dir

Delta function so it's just a spike at one

location that might not even be learnable and then it's just like it's here and even if you got even if you have a spike at 0. five and you it just then it doesn't matter what you observe because there's no probability Mass on outside of that one Spike so it's like those observations are are like discarded whereas the uniform

prior um is which is kind of like the
frequentist case
interestingly is like the the I mean
it's the
least bias in terms of its
sensitivity but then then you realize oh
but then the but the updating is the
bias that is statistical
bias right cool thanks a lot and if
anyone else is interested in like
building these kind of tools or
something please reach out to me because
I think it' be pretty nice to have some
more visualizations of how these things
play
together so but thanks a lot for the
explanation cool yeah it'd be so fun
like for each equation just like even to
like we you could have like natural
language and
translations pseudo
code some different languages some
interactables like there's what are the
kind of like different
forms that that we could interact with
and then also
like where and when and how like in
theory and in the specific
implementations you know the dozens of
implementations like where did the
modelers choose to use the energy minus
entropy kind of statistical
mechanics decomposition of f where did
they use a complexity minus
accuracy kind of Statistics basing
information Criterion type and then
where is there like a Divergence minus
evidence

like if they're
equivalent then does it matter or like
are one of these more computationally
intensive to compute or is there like is
there a reason or a setting why one of
these decompositions on F is better and
then that really gets opened up in the
policy selection for expected free
energy like we talk the most about
epistemic plus pragmatic value that's a
very interesting decomposition for for
policy
selection but it's equivalent to these
two ambiguity plus
risk kind of classic control theory
esque um investing type
decompositions and then both of these
are equivalent to the again energy minus
entropy but then it's like but then are
we even talking about planning if we're
talking about energy minus entropy or
how how and then
like what if we had like a vent diagram
it's like here's the total
amounts and here's the the variance
that's getting partitioned into
epistemic here's the variance that's
getting partitioned into pragmatic now I
don't think the
same this I don't think the same
variance is going to ambiguity and risk
nor to energy entropy because the terms
are not equivalent to epistemic plus
pragmatic value so it's like so what
loads on pragmatic value then does that
get
does that get you know loaded onto risk
or onto

ambiguity does pragmatic value get loaded onto entropy or onto energy again is there a reason or how did different modelers do it the top line for G is is the most common and the top line for for f is the most common I don't know though but those are like they're they're given as a first line because they're kind of like the classic form but it's it's an interesting topic the rule of four or more octopus yeah what is the rule the rule of four is an idea from teaching math that developed in the 90s that was like every concept you look at you try to have at least four different representations usually they're verbal numeric symbolic and um verbal numeric symbolic and visual and uh what what you were saying about like we have all these different representations and we could add more was just echoed that for me cool yeah and here we have we have there's we have different words in Lang different natural languages different graphical um symb it could include executable code and then there could be some some wild cards games and stuff like that but yeah it always pops up it seems it be I mean for each of the equations we could have a column for these and then that creates like you know if there's 50 equations then it's just like we can

just say okay here's the empty cells

let's try to get something into each of
these cells

CS yeah

that's I'll just delete this because

SE

um yeah I mean this is cool and these
are great questions that's what chapter
2 again we haven't even gotten to active
inference this is saying okay op um Bas
optimality

probabilistic probab probably

approximate basee like pack um uh

probably approximately

correct here's the par partition

splitting systems into these four nodes

minimum of four that's another rule of

four or more kind of splitting into four

two blanket States internal and

external getting into what's basically a
variational auto encoder here it is is

framed in the Sens making

setting and then extending that

variational autoencoder into the

decision- making

setting showing that that expected free
energy unified objective or or

imperative or kind of like loss function

a lot in common with with loss functions

that the special cases of expected free
energy fall out as other

well-known pis stics like if you only

consider pragmatic value you get

expected utility Theory no epistemic

value nothing to learn special case of G

expected

utility in contrast if all outcomes are

equally preferable so there's no

pragmatic value on the table then
epistemic
value yields
Infomax so it's kind of like
saying if you're going to compose with
these basy in
Legos either in the the real time
inference and then also extending out
into the the planning and the action
selection like anything any any any
brick structures you build on the low
road are going to you're going to be
able to approximate them with
variational basian
inference chapter 3 is going to come in
with with more of this phys
flow based
model and and say and um for the
organism like observing itself or for
some for you observing something
else something will act as if it is
bounding its surprise about something
else so low roads like saying anything
you build you're going to be able to
approximate it do variational inference
with that surprise
bounding and then here is more of the
the physics saying and you can model
things such that it that it looks like
that's what they're
doing and then in
four that's where the generative models
actually get
shown and it's like this is this was
built up on the low road and it's kind
of compatible from the high road and
that's why it's placed at the
middle it's just that also four has like

again some more mathematical
background like Jensen's inequality and
properties
logarithms um variables and their
dependencies in graphical
models but then figure 4.3 does bring us
to the discrete and the continuous time
generative models
so that's why it's always just kind of
like like many codes of paint like just
kind of reading through because there's
not really just one order especially as
everyone's going to have different like
questions about different parts um so
that's kind of fun like what are the
more the more pure background
materials like putting this as figure
4.1 might make it seem like this is
like finally we got to act in it's
figure 4.1 it's like well no maybe that
that's just a generic property of
logarithms cool um
well so uh next week for
S we uh it'll be
later
or join at this time next week and uh
either Andrew or I
will revisit applying active again
there's no there's no specific order so
that that can be helpful like chapter 7
goes into essentially the top half of
figure 4.3 so may be useful to to see it
and um we can go into
pmdp python based very imperative pseudo
code like form uh format for the
programming like get the observation
update this do that calculate this do
that um or we could explore a little bit

towards RX and fur which has this
declarative style where it's more like
you declare the base
graph full stop then you declare a kind
of like logistical scheduling of how how
do you update different parts of it so
that has there's a few more steps that
are possibly initially
unintuitive but then in operation it
gives the RX infer style of declarative
programming a lot more performance and
flexibility
okay cool any I mean any with anything
else in the times between just make make
the notes on the chapter Pages or just
prepare them
however oh I'll copy and Dave's on
note ruer four in toolic linguistics
they don't write abstracts like they
used
to okay thank you see you all later bye